How SPROUT was built

1. Purpose and thanks

SPROUT (Simulated Processes for Reasoning about Outcomes Under Treatments; in public, "a browser-based crop and soil simulator for teaching") is a daily-step crop, soil water, carbon and nitrogen model written in Python. It runs in a web browser through Pyodide, so a student can run 38 years of a crop at a chosen site with no installation. It simulates corn, silage corn, sorghum, soybean and winter wheat, with corn and soybean rotations, a cereal rye cover crop, tillage, irrigation, split nitrogen and manure.

SPROUT exists because of Cycles. Its purpose is to let students work with a model that behaves the way Cycles does, so we took Cycles as the reference throughout. We built it independently from the Cycles paper and supplement (Kemanian et al. 2024), the Cycles input and output documentation, and published science, and we used runs of Cycles v1.5.20 as the test of the result. In a small number of places the published sources were silent and we measured a constant on Cycles output. This document lists each of those places (Section 6), so that they can be replaced with documented values, and it records where our reading of a printed equation differs from what we implemented (Section 5), so that you can tell us if we have misread it.

Everything below was checked against the code and the data in the repository. Where something could not be checked, it says so.

2. How the build was organised

Provenance classes. Every constant and function in the engine is recorded in a ledger (`agricultural/PROVENANCE_LEDGER.md`, with a script that fails if any is missing), with a class:

- A: form and values documented in the paper, SI, Cycles documentation or a Cycles data file. We also use A for a few exact constants of Cycles that we identified with designed runs (snow, forage fraction).
- B: form and values from another published source.
- C: a documented or published form whose parameters we fitted to Cycles output.
- D: form and parameters both fitted to Cycles output, because no source gave them.
- E: our own assumption, convention or placeholder.
- O: a published form with one or two constants fitted to observations rather than to Cycles output.

The target was to leave nothing in class C or D. We have not reached it, and Section 6 lists every remaining item that depends on Cycles output, so the list is complete rather than a count.

Order of work. Each process was built from the sources first and compared with Cycles second. Where a quantity could only be set from Cycles, it is labelled and listed.

Adoption rule. A documented law replaces another if, at every nitrogen rate, pooled correlation with Cycles stays above 0.8 and the grain error rises by no more than about 0.05 Mg/ha. A change fitted to Cycles must

show an improvement whose paired-bootstrap 95 percent interval on held-out sites excludes zero.

Test sites. Development used a 24-site quick set and 94 sites (`sitesets/fresh94.json`). The 94 sites were used repeatedly to choose between options, and some constants measured on Cycles output came from runs on development sites, so they are a development set and not a clean test. A further 100 sites (`sitesets/holdout.json`) were locked until 2026-10-30, when they were run once on the final engine (Section 7). The engine was frozen at that commit; the only code changes made during the run were guards that let the harness skip a site where Cycles harvests no crop.

How it was built and checked. SPROUT was developed by a single author working with an AI coding assistant (Claude Code). Several of the subject areas are outside the author's own expertise, so the process was built so that nothing was accepted on the assistant's word or on trust. Every claim in this paper was checked against code, data or a source document before it was written, and a standing rule was that a number is treated as stale until it is re-read from a run. This practice also found mistakes, which are part of the record: a stale nitrogen-loss comparison, a soybean result read from a run that had not finished, a curve adopted on a secondhand report of a paper that, when read, said the opposite, and a harvest index slope first treated as fitted that turned out to be the documented formula. Each was corrected in the repository and the paper, and each is dated in the plan.

Scale of the work and how it was tested. SPROUT was not produced by transcribing the Cycles equations once. Development ran from 2026-09-21 to 2026-10-29, with 690 commits to the agricultural directory (778 in the repository) on 31 working days. The engine is about 5,100 lines of Python, with 93 functions and over 200 module constants, all recorded in the provenance ledger. The record of the work is kept in the repository: 240 dated entries in a 1,272-line rebuild plan, a 4,561-line mechanism log, a refit plan, 39 stored scorecards and 251 scripts in the prototype folder, many of them single-purpose tests. Earlier decisions to 2026-10-17 are archived in a further 691 KB file.

The tests were of several kinds, and most changes had to pass more than one of them:

- *Multi-site comparison with Cycles.* Every full scorecard is 94 sites by 38 years at three nitrogen rates for corn, silage, sorghum and soybean, and 60 sites for winter wheat, about 50,000 simulated site-years in each model. A keyword count of the plan finds 108 entries that report a 94-site comparison, 102 that report a 24-site quick comparison and 18 that report the 60-site wheat comparison.
- *Attribution with forced states.* To find which process caused a gap, Cycles' own daily canopy, water or nitrogen state was forced into SPROUT, one at a time and together. About 60 plan entries use this kind of attribution.
- *Designed Cycles runs.* To identify exact constants and structures without fitting, we changed one input of Cycles at a time (for example the N concentration parameters, planting density, maximum soil coverage, residue removal, snow inputs) and read the response. This found the forage fraction, the snow constants, the harvest index slope, and the dependence of soybean nitrogen on planting density and of canopy interception on soil coverage.
- *Paired and held-out statistics.* Changes fitted to Cycles were accepted only if a paired bootstrap on held-out sites excluded zero, and documented laws if grain error stayed within about 0.05 Mg/ha; 47 plan

entries report held-out or bootstrap results. The county-yield comparison followed a protocol written before it was run.

- *Independent observations.* USDA county yields at 94 sites for 35 years, AmeriFlux tower biomass, leaf area and evapotranspiration at three Nebraska sites, and published field data were used as checks and, where they disagreed with Cycles, as the arbiter.
- *Guards against regression.* A parity check confirms that the engine the simulator page runs gives identical output (difference 0.0) to the engine used in the scorecards, a snapshot check detects unintended changes, and a script checks that every constant and function is covered by the ledger.

Many of the tests found nothing to adopt. By the same keyword count, about 100 of the 240 plan entries record a change, option or published alternative that was tried and not adopted or was found to be worse, including several denitrification laws, two root distributions, a clay-dependent microbial efficiency, a fitted evaporation law and a late-season growth switch. Those records are in the repository so that they can be examined.

3. Inputs and the reference runs

Weather is the gridded daily record for 1980 to 2016 at the nearest 0.5 degree cell and soil is the nearest STATSGO2 profile. The Cycles reference runs use v1.5.20 with its hourly finite-difference hydrology option (HYDROLOGY_OPTION 1) and the curve number runoff method. Operation files set STANDING_RESIDUE_AT_HARVEST to 50 percent for corn, soybean and silage and 40 for wheat, RESIDUE_REMOVED to 0 for grain crops and MAXIMUM_SOIL_COVERAGE to 100. Cycles daily rows are end-of-day states, so in the comparison a day's flux in SPROUT acts on the previous row's state.

4. Process by process

Each subsection says what the sources state, what the engine does, the class, and where Cycles output or observations were used.

4.1 Evapotranspiration, runoff and soil water

Implemented as documented (class A): FAO-56 Penman-Monteith (agreement with Cycles reference ET within 0.4 percent, correlation 1.000); curve number runoff (SI.1 to SI.6, see Section 5 for our reading of two printed signs); Campbell's hydraulic conductivity and the sub-daily redistribution with travel time (paper Eqs. 1 and 2); the transpiration and water use efficiency growth terms. Root water uptake follows Campbell 1985 and CropSyst (class B). Soil hydraulic properties follow Saxton and Rawls 2006 (class B), with one option, the air-entry potential anchor, chosen by yield agreement with Cycles (class C).

- **Soil evaporation (class O).** The paper divides evaporative demand among canopy, residue, snow and soil using radiation interception but gives no formula for the soil term. The form and three constants were first measured on Cycles output, then tested against USDA NASS county yields, which we treat as the arbiter where they disagree with Cycles. Of six settings tested, the adopted law gave the best rainfed eastern corn (level 1.00 of observed, within-site r 0.50); the FAO-56 two-stage form gave 0.92 and 0.40. A law fitted to Mead tower evapotranspiration was tried and not adopted.

- **Residue cover and position (class A structure, class B cover law, one class E choice).** Surface residue is split into standing and flat. The standing share at harvest is the operation-file input; a tillage pass flattens the square root of its mixing efficiency of the standing residue (Cycles input guide). Ground cover is Fc = 1 - exp(-K M) (Gregory 1982). K is 0.40 per Mg/ha for corn, the upper end of the range we found for maize (about 0.30 in the WEPS documentation) and close to the 0.384 we saw in Cycles output; soybean uses the same value as an assumption (class E). Standing and flat residue decay at the same rate, which is what we measured in Cycles output (ratio 1.13).
- **Residue water store (class A structure, class B capacity).** The paper says rain wets residue first and that residue takes its share of evaporative demand. The engine holds a water store in the residue, filled by rain up to a capacity and emptied by reference evapotranspiration times cover times one minus canopy. The capacity is 3.10 kg water per kg, the maximal water content of maize stem particles (5 cm, 20 C) in Iqbal et al. 2013 (Table 2; wheat 2.33, soybean 2.16). This is an immersion value, so it is an upper bound for field residue.
- **Snow (class A by behaviour).** Rain and snow split at 1.0 C, degree-day melt over a triangular daily temperature wave (melt base -1.0 C, 2.5 mm per C) and sublimation equal to reference ET. The constants are those of Cycles, identified by designed runs that reproduce its snow water to 0.1 mm over 100 cases.
- **Moisture adjustment of the curve number (SI.7).** Described in words in the SI. We used the depth weighting in Williams et al. 2011 (partly class E).
- **Root distribution (class E).** Not given in the sources we have; exponential density with a 0.3 m scale. A CropSyst-style linear taper was built and tested and was worse against Cycles and against county yields.
- **Soil temperature.** Layered implicit heat conduction with a damping depth inside the range quoted in the SI (class B form, parameter inside that range). It has no snow or residue insulation, so winter soil is about 1.5 C colder than Cycles'. A simple lag filter is used only to set planting dates (class E).

4.2 Crop growth

Implemented as documented (class A): growth is the lesser of a radiation-limited and a transpiration-limited term (paper Eqs. 3 to 5) with crop-file efficiencies at a growth scalar of 1.0; the double-sigmoid canopy (Eq. 6) with the plant density adjustment (Eq. 7); thermal time, vernalization and photoperiod from the crop file; the shoot partition (SI Eq. 8); the planting window from the operation file.

Three things in the implementation go beyond the printed equations, and we list them so they are visible: radiation-limited growth is multiplied by a temperature ramp (the same ramp as transpiration unless the crop sets its own); the canopy used for growth on a given day is the previous day's canopy; and nitrogen stress (a nitrogen nutrition index) multiplies growth as well as slowing canopy expansion. The paper states that the rate of canopy change is reduced by N stress; we did not find a statement about the other points, so they are our choices (class E).

- **Harvest index (class A).** The guide gives HI = Hmax - (Hmax - Hmin) exp(-Hk fG) with Hk = m (1 - Hmin) / (Hmax - Hmin). With the crop-file values this gives 1.3077 for corn, silage and sorghum, 3.4 for soybean and 2.5 for winter wheat.

- **Shoot partition mid-point TTf50 (class A by behaviour for corn, silage and sorghum).** SI Eq. 8 needs TTf50 and we found no printed value for it. We read it from designed runs of Cycles as a single scalar of the printed form (exponent 4): 0.417 for corn, 0.412 for sorghum and 0.411 for soybean, so the engine's shared 0.42 agrees to within 0.01. Thermal time was normalised by the season's maximum (1834 against the crop file's 1800), which biases the read by about 0.01. Winter wheat's 0.32 was not read this way. We then tested it on measured above-ground biomass at the Mead towers (120 points, June to September): the squared relative error is 0.0264 at 0.30, 0.0103 at 0.42, 0.0115 at 0.50 and 0.0162 at 0.56, so the observations bracket 0.42 to 0.50. Because the original value came from Cycles, it is also listed in Section 6.
- **Canopy plateau (class A by behaviour, one constant for all crops).** With epsilon_ix = 1 and the paper's constants, Eq. 6 peaks near 0.99, but designed runs of Cycles show that the peak radiation interception of corn, silage, sorghum, soybean and winter wheat is 0.95 times MAXIMUM_SOIL_COVERAGE (0.949 to 0.954 in every case; with the setting at 90 percent the peak is 0.86 to 0.857). SPROUT uses 0.95 for every crop. Earlier versions used a separate plateau per crop (0.951, 0.924, 0.946, 0.949 and 0.853) measured on Cycles' unstressed days, which differ because the realised annual peak varies; replacing them with 0.95 changed corn, silage and sorghum by 0.01 Mg/ha or less, soybean grain error by +0.03 and winter wheat by +0.006, +0.011 and -0.018 Mg/ha at N = 0, 90 and 150. For reference, maximum interception measured at a Nebraska site (Kukal and Irmak 2020, Table 2) is 0.92 and 0.97 for short and long season maize, 0.99 for soybean, 1.00 for sorghum and 0.97 for winter wheat.
- **Canopy shape (class A by behaviour, since 2026-10-31).** Designed runs of Cycles under constant, unstressed weather show that the canopy of corn, silage, sorghum and soybean follows Eq. 6 with a = 6, b = -20, c = -16 and d = 15 (the paper prints c = -15 and d = 16) on a single axis TTf = thermal time / (MATURITY_TT - THERMAL_TIME_TO_EMERGENCE), evaluated half a day before the reported end-of-day value. With those constants and the plateau as the only free parameter the fit has rms 0.0014 to 0.003 in interception for the four crops; the printed c and d on the same axis give 0.14. The fit is the same at 18, 22 and 26 C, at MATURITY_TT of 1500, 1800 and 2100, at emergence thermal times of 40, 65 and 100 and for two planting dates. The senescence midpoint (-c/d) is therefore at 1.067 of that axis, not 0.9375. This agrees with the shift of 0.12 that we had fitted to measured leaf area at the Mead towers (next item). The stress response is also structural: the paper says the first derivative of Eq. 6 is the rate of change and that the rate is reduced by N stress. In Cycles a stressed season keeps a low canopy level afterwards (it does not catch up), and reproducing Cycles' daily canopy from its own stress series requires the stress to multiply only the rising part of the derivative. SPROUT now integrates the derivative daily this way, with a linear N stress factor (as the paper says) and a water stress factor raised to the power 2.0 for corn, silage and soybean, 1.0 for sorghum (the paper names only N; in Cycles water stress also reduces the rate, CropSyst applies water stress to leaf area production, and the exponent is fitted to Cycles' behaviour, class C: on 94 sites 2.0 improved soybean by 0.024 Mg/ha [0.013, 0.035] and left corn and silage unchanged, while for sorghum it cost 0.036 [0.005, 0.073] at N = 400 on 24 sites; an exponent of 1.0 was the neutral first choice). What we have not found is the exact map from Cycles' reported stress to its rate multiplier: measured ratios of realised to unstressed canopy increments are 0.96 / 0.85 / 0.64 / 0.41 / 0.11 at N stress 0.075 / 0.15 / 0.25 / 0.35 / 0.6, steeper than linear, and N and water together act more strongly than their product. On 94 sites against the previous fitted shapes the paired grain error changed at N = 0 / 150 / 400 by -0.022 / +0.060 / +0.036 Mg/ha for corn, +0.023 / +0.034 /

+0.098 for silage, -0.051 / +0.041 / +0.100 for sorghum and +0.081 for soybean (positive is better; the soybean, silage N = 400, sorghum N = 150 and 400 and corn N = 150 intervals exclude zero). Winter wheat and the rye cover crop keep the earlier form.

- **Soybean biomass.** After the planting window was corrected, soybean thermal time matched Cycles' and the remaining excess was early canopy expansion and growth under water stress: above-ground biomass 11 to 14 percent high through the season, with Cycles' soybean showing no N stress at all. The water exponent above brought the soybean bias to +2.6 percent. Our earlier statement that soybean biomass was 13 percent high with harvest index 11 percent low was an artefact: the comparison used total biomass including roots. On above-ground biomass the model is 3 percent high, harvest index (grain over above-ground biomass) is 0.359 against 0.364, and the grain bias is about +3 percent.
- **Planting window (class A input, since 2026-10-31).** The engine planted every crop in the corn window (day 110 to 131). Cycles' soybean operation file uses 120 to 140, and soybean was planted 9.7 days earlier on average in SPROUT (115.6 against 125.3), which put its thermal time and canopy ahead and its biomass about 20 percent high. With the file's window the mean planting day is 123.8, the soybean grain error fell by 0.038 Mg/ha (95 percent interval 0.023 to 0.052 over 94 sites), and the bias went from +7.5 to +5.2 percent. Biomass remains about 18 percent high and harvest index about 11 percent low for soybean, compensating.
- **Early-season N uptake of corn (class E with a sourced estimate, since 2026-10-31).** At no nitrogen the model held more soil mineral N than Cycles and still took up about 30 percent less in the first third of the season, so N stress was too strong early. The supply limit follows Godwin and Jones (1991) with root length from root mass; the root mass at emergence was a 10 kg/ha assumption. We replaced it for corn and silage corn by 25 kg/ha, the weight of the seed (about 0.3 g at about 79,000 per ha is 24 kg/ha), because CERES starts the root from seed weight. On 94 sites the paired grain error at N = 0 / 150 / 400 improved by 0.084 [0.049, 0.128] / 0.005 / 0.000 Mg/ha for corn and 0.036 [0.015, 0.060] / 0.014 / 0.005 for silage. Sorghum was inheriting the same 25 kg/ha from the corn configuration without our noticing; making it explicit, 25 against 10 improves sorghum by 0.082 [0.051, 0.118] / 0.014 [0.004, 0.028] / 0.000, but sorghum seed would give about 5 kg/ha and 5 was worse, so the parameter behaves as early root development and its sorghum value is a fitted choice (class C). A specific root length of 150 m/g instead of 100 (literature 50 to 150) improved a 24-site test further but is a choice we made with Cycles in view, so it is not adopted and is listed as a suspect.
- **Senescence timing of maize (class O, now superseded for the four crops above).** Before the canopy law above was found, the senescence term was moved later by 0.12 in thermal-time fraction, with the senescence clock accelerated by water and nitrogen stress, fitted to measured leaf area at the three Mead towers (maize 2001 to 2007, 80 points) and checked leaving one year and one site out. The shift of 0.12 is what the corrected constants give (1.067 minus 0.9375, scaled by the axis), so the tower data and the Cycles runs agree independently.
- **CO2 effect (class A form, class O exponents).** The paper says efficiencies respond to CO2. The exponents are from free-air enrichment syntheses (Ainsworth and Long 2021): C3 crops 0.38 on both efficiencies (about +14 percent at 550 ppm), C4 crops 0.45 on water use efficiency and none on radiation use efficiency.
- **Root growth leaving the plant (class C).** Cycles output implies that about 40 percent of root growth does not stay in root biomass; this is an inference from the output, not a documented constant. The

engine retains 0.6. Tests of 0.5 and 0.7 changed pooled correlation by less than 0.006.

- **Winter wheat clock (class A).** The thermal-time fraction is thermal time since planting over thermal time at maturity. For winter wheat the season-end thermal time in Cycles' output ranges from 1183 to 3226 degree days across 814 site-years (mean 1858), so the engine runs each season twice, once to find the season's own end and once to use it. This was the largest single gain for wheat.
- **Growth efficiency of wheat (open).** Unstressed wheat in SPROUT makes about 14 percent more biomass per unit of intercepted radiation than Cycles. A growth scalar of 0.95 is kept for wheat as a labelled suspect; removing it makes yield worse. We have not found the cause.
- **Silage harvest (class A by behaviour).** Harvest at 85 percent of maturity (operation file). Forage yield is 0.95 times above-ground biomass minus root: harvest.csv reports this ratio to within 0.0005 in 111 harvest rows at three sites, and designed runs changing RESIDUE_REMOVED and STANDING_RESIDUE_AT_HARVEST left it unchanged. We have not compared it with real stubble heights.
- **Cold damage (class E).** The documentation describes a cold damage factor that depends on temperature. We implemented a single-threshold rule (a night below the crop's cold damage threshold after flowering ends the crop). We did not implement the graded factor.

4.3 Nitrogen in the plant

- **Demand (class B).** The critical dilution curve of Lemaire, Jeuffroy and Gastal 2008 (corn and silage 3.4 and 0.37, sorghum 3.9 and 0.39, winter wheat 5.3 and 0.44), with the new-growth concentration read at the biomass reached without N stress (paper Sec 2.5) and luxury uptake allowed. An earlier scale on demand (0.70 to 0.75), measured on Cycles, was removed for corn, silage, sorghum and wheat.
- **Stress (class B form, class E constant).** Growth is proportional to the nitrogen nutrition index above a floor of 0.2 of the critical curve (the linear relation of relative growth rate to plant N concentration, Ingestad 1979 as cited by Lemaire et al. 2008). The floor is an assumption; we chose 0.2 because Cycles corn never dilutes below about that ratio, so it is not independent of Cycles.
- **Supply limit on uptake (class B form, class C exponent).** The paper says water flux determines access to mineral N and gives no equation. The engine uses the Godwin and Jones 1991 availability function (1 - exp(-kC), root length density times a soil water factor times 0.006 kg N/ha per cm of root). The water factor exponent is 1 where their text gives 2; we set it to 1 because 2 reduced yield too much when tested against Cycles. Doubling the 0.006 constant changes yield by under 1 percent.
- **Uptake and growth solved together (class B).** CropSyst 1994 Eqs. 29 and 30.
- **Legume path for soybean (class A mechanism, class B constants, one class C).** Soil N is taken first, fixation fills the unmet demand, and residue N comes from plant N (paper Sec 2.5). Demand follows the soybean shoot N dilution curve measured by Divito et al. 2016 (%N = 3.53 W^-0.08, W in Mg/ha, from 1 Mg/ha to R5; the fit is weak, R2 0.15, and we extend it to maturity). The authors note that soybean's curve is much flatter than in other crops. Grain carries 0.75 of whole-plant N, about 0.80 of above-ground N, within the published range of N harvest indices (0.44 to 0.91, Tamagno et al. 2017, abstract only). The root share of plant N (0.065) was read from Cycles' harvest file (class C). Ordonez et al. 2020 give a root-to-shoot N ratio of 0.13 for soybean, i.e. a root share of 0.115; with it, soybean yield is unchanged but corn after soybean at zero N falls from 0.79 to 0.57 of Cycles' (grain error 0.92 to 1.36

Mg/ha), because less N returns with the soybean residue than Cycles returns. We kept the Cycles-read value so that rotations behave like Cycles, and list it in Section 6. These published values put more N in SPROUT's soybean than Cycles carries: plant N is 344 kg/ha against 203, and fixation is 302 against 142 (85 percent of plant N, against 50 to 60 percent in Salvagiotti et al. 2008). Designed Cycles runs explain part of this: Cycles' above-ground soybean N is linear in N_MAX_CONCENTRATION, scales with DENSITY raised to the dilution slope, does not respond to RUE or TUE, and averages 0.76 of the crop-file curve (0.69 for corn), with seed N of only 3.9 percent. We first tried Lemaire's C3 non-legume critical curve by analogy, which matched Cycles' plant N, but that rested on a secondhand report of Divito et al. that turned out to be wrong, so we dropped it. In a corn-soybean rotation on 24 sites, corn after soybean is 0.79 and 0.94 of Cycles' at N = 0 and 150. This is question 5 in Section 9.

- **Other nitrogen constants read from Cycles (class C or D).** The share of plant N in the above-ground part (0.88) and the root N carried out with lost roots (0.12 of plant N for corn, 0.082 for wheat) were measured on Cycles. The grain share of above-ground N (0.70 for corn and sorghum, 0.75 for wheat) is a literature harvest index that we have not re-verified. The grain and straw N concentration fields of the crop file are not used.

4.4 Soil carbon, nitrogen and decomposition

The six-pool structure with first-order residue decay (SI Eqs. 10 to 14), the saturation factors fH and fD (SI.12 and SI.13), the saturation capacity of soil carbon as a function of clay (Hassink and Whitmore, as in Pravia et al. 2019), the tillage speed-up of decomposition and the 2 percent per day decay of its disturbance, and the moisture response as a Hill function with half-response 0.22 and power 3 (Cycles documentation) are implemented (class A and B).

- **Microbial carbon use efficiency (class A).** The SI gives 0.33 to 0.44. The engine uses 0.44 times the saturation factor fH. An earlier clay-dependent fit to Cycles was removed because it was neutral.
- **Saturation factor fH (class A).** SI Eq. SI.12 as printed: exponent 6 and no floor. An earlier version used an exponent of 12 and a floor of 0.2, fitted to Cycles' humification near saturation; the printed form is neutral on the current engine (60 wheat sites MAE 0.377 / 0.654 / 0.799 against 0.379 / 0.655 / 0.797 at N = 0 / 90 / 150; corn, silage, sorghum and soybean within 0.02 Mg/ha on 94 sites).
- **Per-source efficiency and the aging factor fA.** The SI's per-source retention fractions (fra, frt, frz, frm) are folded into a single efficiency, and fA is set to 1.0 (a placeholder, class E).
- **Decay rates (mixed).** Surface residue decays as in CERES (Godwin and Jones 1991): 20 percent carbohydrate, 70 percent cellulose and 10 percent lignin at 0.8, 0.05 and 0.0095 per day times the environmental factor (class B). We note that the chapter's text gives 0.8 per day for carbohydrate while the Fortran listing printed in it uses 0.2; we used the text. The WEPS law of Steiner et al. 1999 is kept as an option and still governs buried root residue. The base rate of soil organic matter turnover is 0.00015 per day (Pravia et al. 2019, class B). The microbial turnover rate and the manure rate were back-calculated from Cycles output (class C), and the temperature optimum of the decomposition factor (37.5 C in the Campbell function of Stockle and Campbell 1989) was fitted to Cycles' composite factor (class C). The CERES law replaced WEPS on 2026-10-29: on 94 corn sites the paired grain error improved by 0.20 and 0.42 Mg/ha at N = 150 and 400 (intervals exclude zero) and worsened by 0.06 at N = 0, with

silage and wheat neutral to slightly better. We have not yet checked the resulting residue cover against field data.

- **Pool carbon to nitrogen ratios.** Soil organic matter and microbes use 10.0 (the Cycles guide gives 10:1 for soil organic matter; for microbes the SI gives a range, see below). Residue N is the plant's above-ground N minus grain N. Root residue uses a C:N of 52, measured on Cycles output (class C). Field measurements at peak biomass in Iowa give 30 for maize roots and 20 for soybean roots (Ordonez et al. 2020), and they note that C:N rises as roots senesce. Using 30 in SPROUT raised the grain error at N = 150 by 0.15 Mg/ha and leaching to 1.65 times Cycles' on 24 corn sites, so we kept 52 and list it in Section 6. A separate manure-residue C:N of 30 is carried in the code but unused: manure's organic fraction joins the surface residue pool with the C:N of its source in the fertilizer catalog. The fallback residue C:N of 86 now applies only to silage residue (class D); the other crops take residue N from plant N.
- **Variable microbial C:N (SI.14).** Implemented as a switch (on). We could not reconcile the printed mineral-N factor with the stated range (maximum 8.5 + 5.5 x 0.5 = 11.25, stated 14), and the printed factor increases with mineral N whereas we expected the opposite. We implemented the factor 0.5 / (1 + (N/8)^3), which decreases with mineral N; a switch for the printed form is in the code. Both reproduce the SI worked example (8.65 at N = 8 ppm). This is question 4 in Section 9.
- **Nitrification (class B).** Johnsson et al. 1987 (SOILN Eq. 9): transfer from ammonium to nitrate at k 0.2 per day, none when the nitrate to ammonium ratio exceeds 8, with SOILN's moisture response and the Stockle and Campbell temperature function.
- **Atmospheric nitrogen deposition (off by default).** An option adds 8 kg N/ha/yr, half ammonium and half nitrate, to the top layer daily. Cycles has none, so the default is zero and a student sees Cycles-like zero-nitrogen yields. We do not give a literature source for the 8 kg here; it is a placeholder for a typical Corn Belt value.

4.5 Nitrogen losses

- **Denitrification (class A form, amplitude 1.0).** The EPIC and APEX law as published (Kemanian and Williams, APEX upgrades Eqs. 1 to 4) with an amplitude of 1.0. An amplitude of 2.5 had been fitted to Cycles and was removed. At 1.0, SPROUT's denitrification is 0.3 to 0.4 times Cycles' for corn at 150 and 400 kg N. Two other published laws were tried with their constants unchanged: CERES (Godwin and Jones 1991 Eqs 15 to 18) matches Cycles' denitrification and leaching within 8 percent for corn at 150 kg N but gives 5.9 times Cycles' denitrification with no fertilizer; SOILN (Johnsson et al. 1987 Eqs 15 and 16) gives 0.3 times at high N. Neither is adopted. The Cycles documentation names POT_DENITRIF_RATE and DENITRIF_HALF_RATE only as calibration multipliers, so we could not apply the defaults.
- **Total nitrogen lost.** For corn the sum of leaching, denitrification and volatilization (per year, N = 0 / 150 / 400) is 9.1 / 39.4 / 179.6 kg N/ha in SPROUT and 9.9 / 46.0 / 190.2 in Cycles, so SPROUT loses about 8, 14 and 6 percent less in total. The partition differs more (Section 7).
- **Nitrous oxide (class A and class E).** The share of denitrified N is APEX Eq. 5, with no fitted constant (tested on 149,000 daily states of Cycles: ratio 0.079 against Cycles' 0.081). The share of nitrified N is a constant 0.4 percent. We could not find a published value that fixes this number, so it is an assumption

(class E) that affects only the reported N2O. The emission factor at N = 150 is about 0.65 percent of applied N (IPCC default for EF1 is 1.0).

- **Volatilization (class B form, amplitude calibrated to Cycles, class C).** Daily NH3 loss = Kh F C_TAN / r_a (Genermont and Cellier 1997, Eqs. 3 to 5 and 13; Montes, Rotz and Chaoui 2009 for Ka and Kh; r_a from FAO-56). The soil pH of 7.2 used in this form is a calibration of the amplitude and not a soil property: designed Cycles runs show that Cycles' volatilization does not respond to the soil pH input at all (identical to the digit for pH 5.5, 6.5, 7.5 and 8.5). Designed runs also show that in Cycles the loss rises roughly in proportion to the fertilizer applied (about 14 to 17 percent of it), falls steeply with clay (for one site with the texture changed, 66 kg N/ha/yr at 1.5 percent clay, 30 at 10, 15 at 20 and 7 at 40), also falls with soil water capacity at fixed texture, and depends on weather (the same soil under three weather records gave 27, 60 and 109 kg N/ha/yr), but not on bulk density or soil organic carbon. SPROUT has no clay dependence (across the 94 sites the log-log slope of loss on clay plus one is -0.94 in Cycles and -0.01 in SPROUT), so SPROUT's loss relative to Cycles' is 0.57, 1.20, 1.51 and 2.67 for clay below 8, 8 to 15, 15 to 25 and above 25 percent. The pH of 7.2 puts the mean right and the texture dependence is missing; we have not added a fitted clay term. Further designed runs under constant weather identified much of the structure of Cycles' loss: it is first order in the top-layer ammonium, linear in wind, Arrhenius in temperature (slope 9641 K, matching the product of the ammonium dissociation and ammonia Henry constants), proportional to bulk density over water content, a function of clay alone and of the thickness of the first layer, and reduced by flat surface residue. A trial implementation of this structure raises the across-site correlation of SPROUT's annual loss with Cycles' from 0.32, 0.65 and 0.83 to 0.89, 0.92 and 0.91 at 0, 150 and 400 kg N/ha, but overestimates the level by a factor of 1.5 to 2.7, so at least one factor is still missing. Because tabulated functions of this kind would reproduce Cycles by construction rather than describe the processes, the trial is kept only as a diagnostic and is not part of SPROUT. Field and laboratory studies agree on the direction: ammonia loss from urea falls as clay or cation exchange capacity rises (Martens and Bremner 1989 on 20 Iowa soils, Sunderlage and Cook 2018 on 79 US soils, Pelster et al. 2019 on eastern Canadian soils, a regression of Watson 1990 summarised by Bishop 2011); we have seen these only in summaries, and they are statistical regressions whose magnitudes differ from each other (Pelster et al. report a 90 percent reduction from 10 to 20 percent clay for banded urea, against about 50 percent in Cycles), so none supplies a process equation to adopt. This is question 6 in Section 9.
- **Nitrate transport (class B).** Godwin and Jones 1991 Eq. 1.
- **Application losses (class B).** The IPCC 2006/2019 Tier 1 default fraction (10 percent of applied synthetic N). A weather-dependent equation after Macnack et al. 2013 is in the code but switched off.

4.6 Management

Tillage implements and fertilizer and manure composition are read from the Cycles data files (class A). Irrigation follows the documented automatic-irrigation rule. The cover crop uses the WinterRye crop-file values on the wheat configuration. Dated fertilizer applications are applied on the day given; a dose dated up to 60 days before a spring crop's planting day lands on planting day, and any dose that never meets a growing-season day is reported in the run output.

5. Places where we read a printed equation differently

The following are differences between the printed form and what we implemented, given as our reading. We would value your confirmation, and we have not assumed that any of them is an error rather than our misunderstanding.

1. SI.2 prints (Win - 0.2S)^2 / (Win - 0.8S). In the standard curve number form the denominator is Win + 0.8S. We implemented the standard form. As printed, the denominator is negative for Win between 0.2S and 0.8S.
2. SI.6 prints CN_b / (0.4 - 0.006 CN_b) for the wet-condition curve number. The denominator is zero at CN 66.7 and negative above it. We implemented the form with a plus sign, which keeps CN_wet(100) = 100.
3. Paper Eq. 1 prints the exponent as 2 + 3b where psi_e / psi_s appears, while defining theta = theta_sat (psi_e / psi)^(1/b). Campbell 1985 gives 2 + 3/b. We implemented 2 + 3/b. For b = 5 and psi_e / psi = 0.1 the two exponents give conductivities that differ by about 14 orders of magnitude, so the printed form would not be usable and we assume a typesetting issue.
4. SI.14 reproduces its worked example (8.65), but its maximum, 8.5 + 5.5 x 0.5 = 11.25, differs from the 14 stated in the text. We also do not know the intended direction of the mineral-N term (Section 4.4).
5. SI.10 writes the rhizodeposit term with C_ra where we read C_rz, and the text near it uses f_rm where we read f_rz. We implemented what we think is intended.
6. Eq. 6 prints c = -15 and d = 16, and defines TTf as thermal time over maturity. Designed runs of Cycles match c = -16, d = 15 on the axis thermal time / (MATURITY_TT - THERMAL_TIME_TO_EMERGENCE), half a day back (Section 4.2). The printed pair gives an error ten to a hundred times larger on every crop we tested. We use the values the runs show and ask whether the printed ones are a transposition.

6. Everything that still depends on Cycles output

This is the complete list. Class C items use a documented or published form with parameters set from Cycles; class D items use a form and parameters set from Cycles; class E items are assumptions that were chosen with Cycles in view.

- The winter wheat canopy shape (C) and shoot partition mid-point of 0.32 (O), the linear water factor in the canopy rate (E) and the retained fraction of root growth 0.6 (C).
- The microbial turnover rate (C) and the temperature optimum of the decomposition factor (C). A manure decomposition rate is carried in the code but unused, and three older residue and root rates are superseded by the CERES and WEPS residue laws.
- C:N of root residue, 52 (C, published peak-biomass values are 30 for maize and 20 for soybean), and the fallback C:N of silage residue, 86 (D). The soybean root share of plant N, 0.065 (C, published 0.115). Replacing either with the published value moves results outside our tolerance: the maize C:N of 30 (measured at peak root mass, before senescence) costs 0.15 Mg/ha of grain error at N = 150, and the soybean share of 0.115 lowers corn after soybean at N = 0 to 0.57 of Cycles' yield (0.79 now). We also

derived root C:N from the crop's own N (root N 0.107 of plant N, Ordonez et al. 2020), which gives 68 to 81 and a worse fit (24 corn sites, bias -17 to -28 percent at N = 0, grain error +0.08 and +0.15 Mg/ha at N = 0 and 150). The reason is that the model's root carbon input, 1.5 to 2.0 Mg C/ha a season, is about twice the standing root mass those published values imply (about 0.8). We have not resolved this, and it is likely a larger issue than the two constants.

- Soybean root N share (C); shares of plant N in above-ground tissue (0.88) and in lost roots (C).
- The N stress floor of 0.2 (E, chosen with Cycles in view). We looked for a published value and found none (Lemaire et al. 2008 cites only Ingestad's linear relation of growth rate to N concentration). In Cycles the daily stress is a threshold in the nitrogen nutrition index, not a line, so no floor can be read from it. We tried the nutrition index from both the Lemaire curve and the crop file's own N dilution curve, and the ratio of daily uptake to demand; stress is a threshold with wide scatter in all three.
- The exponent on soil water in the uptake supply term (C), and the air-entry potential anchor of the soil hydraulic option (C).
- The pH used in volatilization (C).
- Winter wheat: a growth scalar of 0.95 (labelled suspect). With the plateau now 0.95, removing the scalar raises grain error by 0.01, 0.04 and 0.08 Mg/ha at N = 0, 90 and 150 (the last two intervals exclude zero), so it still compensates for something we have not found.
- Soil evaporation form and constants (O, first measured on Cycles, then tested on county yields).

How much the Cycles-read constants matter. We varied each one on 24 corn sites (grain yield change at N = 0 / 150 / 400 kg N/ha, with the change in nitrogen losses at 150): a canopy plateau of 0.90 or 1.00 instead of 0.95 changes yield by 1 to 4 percent; TTf50 of 0.32 or 0.52 instead of 0.42 by up to 8 percent, in opposite directions at low and high nitrogen; a corn nitrogen-stress floor of 0.1 or 0.3 instead of 0.2 changes zero-nitrogen yield by +14 and -12 percent and yield at 150 or 400 by 2 percent or less; a 25 percent change in the microbial turnover rate changes yield by 2 percent or less; and the volatilization pH of 6.5 or 7.9 instead of 7.2 changes yield by at most 15 percent at zero nitrogen but changes volatilization by -73 and +165 percent. The soil organic matter turnover base rate (0.00015 per day, from Pravia et al. 2019) is the most influential constant for zero-nitrogen yield (a 25 percent change gives -20 and +18 percent), but it is a published value. The volatilization amplitude (the pH of 7.2, which is not a soil property in Cycles), the nitrogen-stress floor and TTf50 are therefore the Cycles-read constants whose replacement would matter most.

Two exact constants of Cycles that we found with designed runs and may be of use: forage yield is 0.95 times above-ground biomass minus root in every row of harvest.csv, and the snow constants (rain below 1.0 C, melt base -1.0 C, 2.5 mm per C) reproduce Cycles' snow water to 0.1 mm. The engine uses both.

7. How well it agrees

On 94 sites, 38 years each, with continuous crop chains (the engine of 2026-10-31, after the canopy law of Section 4.2; winter wheat was not changed). Pooled correlation of annual grain yield with Cycles v1.5.20 at N = 0 / 150 / 400 kg N/ha (wheat 0 / 90 / 150, 60 sites):

Crop	Pooled r	Bias	Grain MAE (Mg/ha)
Corn	0.909 / 0.892 / 0.944	+4% / +1% / -3%	0.62 / 1.05 / 0.90

Crop	Pooled r	Bias	Grain MAE (Mg/ha)
Silage corn	0.887 / 0.895 / 0.933	+14% / +1% / -1%	1.14 / 1.64 / 1.51
Sorghum	0.887 / 0.859 / 0.903	-13% / -9% / -8%	0.64 / 1.19 / 1.16
Soybean	0.956	+3%	0.34
Winter wheat	0.910 / 0.853 / 0.844	+3% / +6% / +3%	0.38 / 0.67 / 0.78

Held-out sites. The 100 locked sites were run once on 2026-10-30 on the engine as it stood before the canopy law of Section 4.2 (corn, silage, sorghum and soybean have changed since, so the table below describes the earlier engine; the sites cannot be run again without losing their independence, and we have not re-scored them). The run used no change to the engine. One site (b058_440_720) was left out because Cycles harvested no crop there at any nitrogen rate, so 99 sites are scored. Pooled correlation with Cycles (95 percent interval from resampling sites), bias and grain error at N = 0 / 150 / 400 (wheat 0 / 90 / 150):

Crop	Pooled r	Bias	Grain MAE (Mg/ha)
Corn	0.915 / 0.907 / 0.953	-9% / 0% / 0%	0.54 / 0.99 / 0.84
Silage corn	0.904 / 0.898 / 0.939	+6% / +2% / +2%	0.99 / 1.61 / 1.42
Sorghum	0.919 / 0.876 / 0.900	-5% / -7% / -3%	0.51 / 1.10 / 1.08
Soybean	0.952	+14%	0.53
Winter wheat	0.946 / 0.871 / 0.850	-4% / +4% / +2%	0.28 / 0.60 / 0.76

All pooled correlations are above 0.8, so the target holds on sites not used in development. The lowest 95 percent lower bounds are 0.803 (wheat at N = 150), 0.827 (wheat at N = 90), 0.838 (sorghum at N = 150) and 0.843 (corn at N = 0); all others are 0.852 or higher. The point values are higher than on the development sites in every case, by 0.04 or less (wheat was 60 development sites against 99 here). Corn at N = 0 is 9 percent low here against 4 percent on the development sites. Pooled correlation is a weak test on its own. Predicting every year at a site by that site's mean Cycles yield, which uses Cycles' own site means and so is an upper bound for a between-site predictor, already gives 0.93 / 0.83 / 0.86 for corn, and 0.94 / 0.86 / 0.85 for wheat; the model beats it by 0.05 to 0.10 for corn, silage, sorghum and soybean at N of 150 or more, is level with it at N = 0 and for wheat. The informative numbers are the year-to-year agreement within sites: the median within-site correlation with Cycles is 0.72 / 0.84 / 0.94 for corn at N = 0 / 150 / 400 and 0.55 / 0.70 / 0.74 for wheat, and at N = 0 the model's year-to-year spread is 1.5 to 1.7 times Cycles' for corn, silage and sorghum. The holdout is a check against Cycles, not against observations: we have not assembled county yields for these sites. Nitrogen loss ratios (engine over Cycles, mean annual leaching at N = 0 / 150 / 400) are corn 1.09 / 1.23 / 1.27, silage 1.42 / 1.59 / 1.33, sorghum 0.84 / 1.17 / 1.14, soybean 0.94, wheat 1.02 / 1.26 / 1.34.

Pooled correlation does not measure the response to nitrogen. On the last panel of response checks (run before the residue changes of 2026-10-29), the 0 to 150 kg response matched Cycles and the 150 to 400 kg response was 63 to 79 percent of Cycles' for corn, silage and sorghum.

Nitrogen losses relative to Cycles (measured before the canopy change of 2026-10-31 and not re-run; corn, kg N/ha per year at N = 0 / 150 / 400): leaching 4.7 vs 4.6, 15.4 vs 13.3, 108.3 vs 84.8; denitrification 0.9 vs 2.6, 7.0 vs 18.3, 21.3 vs 65.8; volatilization 3.6 vs 2.7, 17.1 vs 14.4, 50.1 vs 39.6; nitrous oxide 0.24 vs 0.50, 1.23 vs 3.56, 4.07 vs 15.74.

Against observations, on NASS county yields (94 sites, 1982 to 2016, protocol in `EVALUATION_PROTOCOL_2026_10_28.md`), re-run on 2026-10-31 with the canopy law of Section 4.2, the within-site correlation of detrended yield is 0.43 for corn against 0.38 for Cycles (difference +0.032, 95 percent interval -0.011 to +0.072), 0.35 against 0.33 for soybean (+0.022 [-0.003, +0.048]), 0.39 against 0.34 for silage (+0.061 [+0.013, +0.122]) and 0.21 against 0.23 for wheat (+0.010 [-0.038, +0.057]). The canopy change moved these by 0.01 or less for corn, soybean and silage; the wheat figure differs from 0.23 in our previous run although the wheat configuration did not change in this step (the earlier county file predates other changes). The half-sample split of the protocol was not re-scored, so the defensible statement stays that the two models are comparable, with SPROUT better only for silage. The mean level is 0.98 of the observed county corn mean (Cycles 0.89), 1.43 for soybean (Cycles 1.31), 1.56 for wheat (Cycles 1.45) and 1.15 for silage (Cycles 1.10). In extreme years (eastern corn) both models exaggerate drought: 1988 observed -23 percent, SPROUT -31, Cycles -43 (SPROUT -35 before the canopy change); 2012 observed -12, SPROUT -36, Cycles -41 (-39 before); neither reproduces 1993 (observed -17, SPROUT +3, Cycles +7).

The Mead flux-tower comparison was last run on 2026-10-28 with settings that have since changed and has not been re-run, so we do not report its numbers here.

For scale, Cycles disagrees with itself by a mean absolute grain error of 0.17 Mg/ha when the soil properties are perturbed by 10 percent at random, 0.74 Mg/ha at N = 150 under a neighbouring weather cell and 1.4 Mg/ha under a neighbouring soil map unit. SPROUT's grain errors against Cycles (0.4 to 1.7 Mg/ha) are of the same order as the last two and larger than the first.

8. Where SPROUT and Cycles still differ

- Denitrification and nitrous oxide are about 0.3 to 0.4 times Cycles' for corn at 150 and 400 kg N (wheat 0.2 to 0.3, soybean 0.24), leaching is 1.0 to 1.3 times and volatilization 1.2 to 1.3 times. Total N lost is 6 to 14 percent lower in SPROUT (Section 4.5).
- Soil water was measured on 2026-10-29 before the CERES residue law: SPROUT was wetter than Cycles in every layer and month (by 0.01 to 0.05 m3/m3), with a surface residue carbon stock 3 to 8 times Cycles', a decomposition factor 1.2 to 1.5 times and winter respiration 2.3 times. The CERES law was adopted to address the stock; water and respiration after it are not re-measured.
- Sorghum runs 6 to 8 percent below Cycles at every N rate.
- Winter wheat: the growth scalar of 0.95 hides a plateau growth efficiency about 14 percent high, and wheat denitrification is 0.2 to 0.3 times Cycles'.
- Soybean and wheat county levels are 1.45 and 1.48 of observed, shared with Cycles.
- Drought years are over-reduced (2012: -39 percent against -12 observed).
- Not yet compared with Cycles as treatment effects: tillage, irrigation, split nitrogen, manure and the planting window; also the rye cover crop and wheat in rotation. In an earlier test (2026-10-23, not re-run) the irrigation response of denitrification had the wrong sign.
- Not implemented: Green-Ampt infiltration, bypass flow, the graded cold damage factor, grazing and multi-species mixtures, the SI's source-specific carbon efficiency multipliers, and use of the crop-file grain and straw N concentration fields.

9. Questions that would help most

1. Why peak interception is 0.95 of MAXIMUM_SOIL_COVERAGE (an Eq. 6 peak near 0.99 would be expected), whether c = -16 and d = 15 and the axis thermal time / (MATURITY_TT - THERMAL_TIME_TO_EMERGENCE) are what Eq. 6 intends, and how reported N and water stress map to the canopy rate.
2. The microbial turnover rate and the temperature optimum of the decomposition factor in the six-pool carbon model (Pravia et al. 2019 gives kx only).
3. The root nitrogen carried out of the plant with lost roots, how much of root growth is retained, the C:N ratios used for root and silage residue, and whether the soybean root share of plant N is meant to be near 0.065.
4. The intended direction of the mineral-N factor in SI.14 and the intended maximum.
5. Whether N_MAX_CONCENTRATION is meant as a maximum curve with realised N near 0.7 to 0.76 of it (Section 4.3), why Cycles' soybean seed N is 3.9 percent, how the grain and straw N concentration fields of the crop file enter the model, and the soybean fixation rules.
6. The form of the ammonia volatilization (what makes the loss fall with clay and soil water capacity, and why it does not respond to the soil pH input), and the defaults behind the calibration multipliers POT_DENITRIF_RATE and DENITRIF_HALF_RATE, and the carbon term of denitrification. With the published APEX law (Rmax 32e-6, half-saturation 60 mg/kg, respiration threshold 50) SPROUT's total N lost is 6 to 14 percent below Cycles' for corn, and it leaches 1.0 to 1.3 times as much and denitrifies 0.3 to 0.4 times as much.
7. Confirmation of our reading of the printed forms in Section 5.

10. Reproducing it

Everything is in the public repository. `rhizo_engine.py` is the engine (the file keeps the name the model had before it was called SPROUT); `engine_api.py` is the interface; `agricultural/PROVENANCE_LEDGER.md` classifies every constant; `agricultural/prototypes/ENGINE_REBUILD_PLAN.md` holds the dated record of every test, including those that failed; `scorecards/` and `sitesets/` hold results and site lists.

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